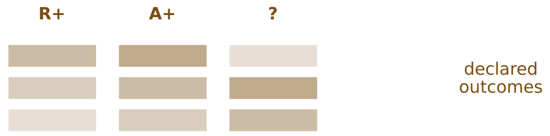
FIG-002. Synthetic truth and named-baseline benchmarking

Stage 10.14 readable review render from the Stage 10.5 panel-evidence crosswalk

A Known-truth regimes for benchmark stress tests

Purpose

Set synthetic residence-positive, amplitude-sufficient, and ambiguous truth regimes before benchmarking.



Reader readout

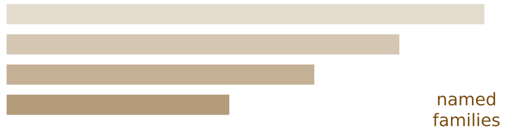
Comparator performance is judged against declared truth structure rather than against a RhoDyn-only score.

Stage 10.2 main benchmark

B Named comparator families

Purpose

Display simple summaries and named external-style baseline families side by side.



Reader readout

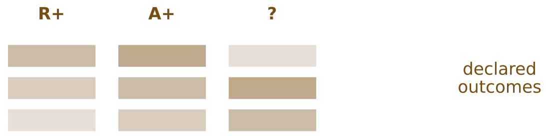
The benchmark includes amplitude, endpoint, AUC, peak, latency, threshold, SciPy peak, scikit-learn, HMM, catch22-style, tsfresh-style, MiniROCKET-style, and ruptures-style comparators.

Stage 10.2 main benchmark

C Accuracy and boundary outcomes

Purpose

Compare RhoDyn decisions against named baseline performance and record where generic features succeed.



Reader readout

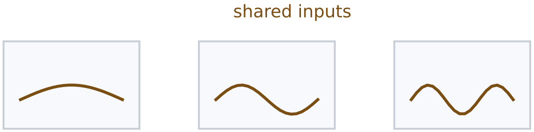
The paper should foreground both RhoDyn-positive regimes and regimes where named comparator families are sufficient.

Stage 10.2 main benchmark

D Public input benchmark summaries

Purpose

Apply the same comparator framing to retained public DRG calcium and ERK GPCR inputs.



Reader readout

The method object and named baselines are evaluated on shared public inputs, not only synthetic fixtures.

Stage 10.2 main benchmark

E Runtime and memory profile

Purpose

Show practical compute cost for the method object and named feature families.

Reader readout

RhoDyn's methodological claim is separable from software maturity, but the benchmark remains computationally inspectable.

Stage 10.2 main benchmark support